



Robin Ko

System Static Timing Analysis

Robin Ko

LSI Logic Corporation



System Static Timing Analysis (SSTA)



Robin Ko

- **Expand the application of PrimeTime to system level perspective**
 - ◆ **Flexibility**
 - ◆ **Scalability**
 - ◆ **Visibility**

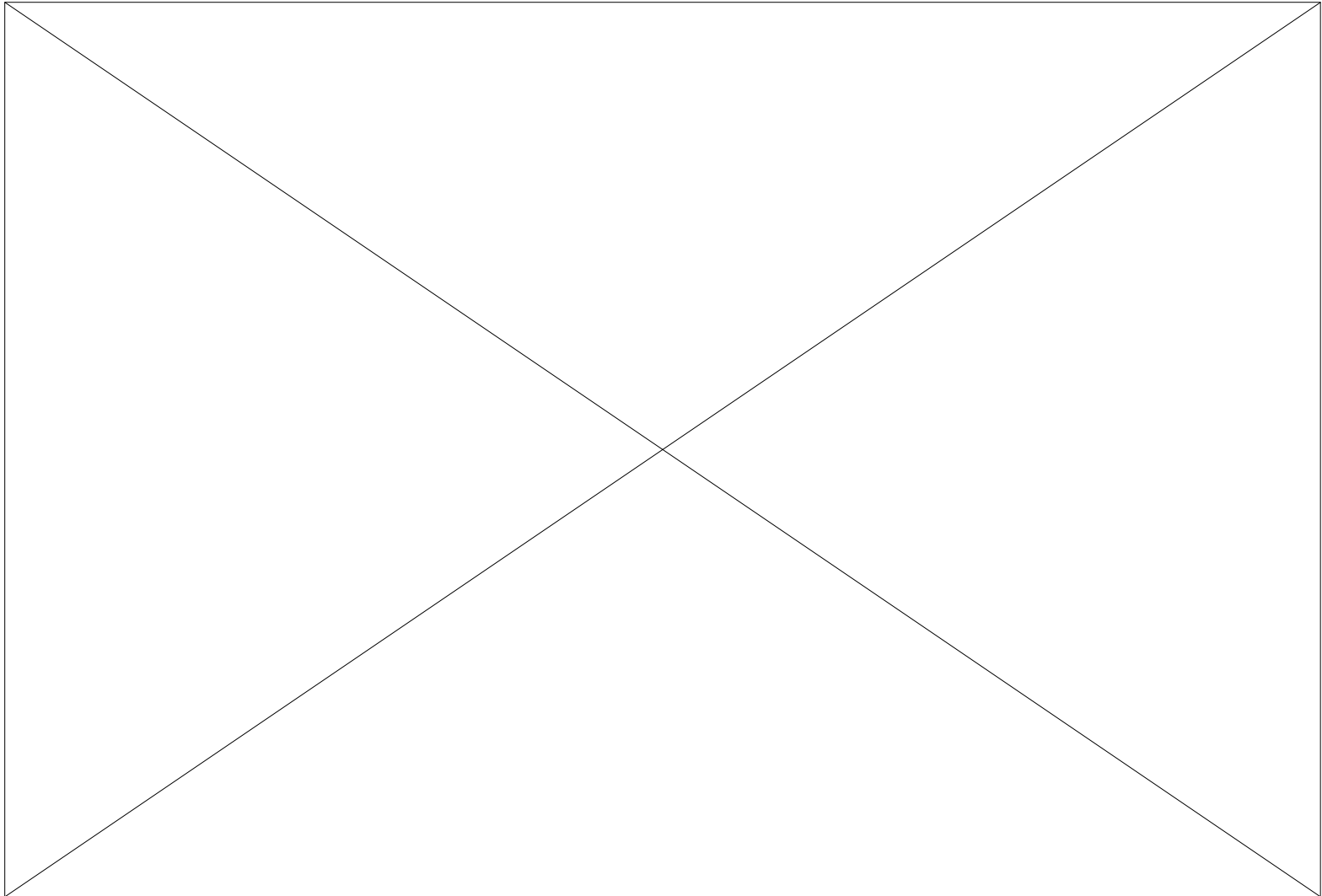
Agenda

- **System static timing analysis (SSTA)**
 - ◆ Main features
 - ◆ Work flow
 - ◆ Benefits
- **Demonstrate the benefits on a DDR interface**
 - ◆ Challenges
 - ◆ SSTA on DDR interface

System Static Timing Analysis

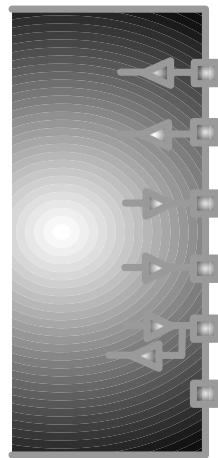
- Main Features
- Work Flow
- Benefits

I/O Static Timing Analysis



I/O Static Timing Analysis

set_input_delay 5 -clock bclk PORT_B_1



0.001 2.45 7.2



SDCLK_O	OUT	PDO08CDG	SDCLK_O	OUT			2.86			3.18		4.43
SDQ_B	IN	PDB08DGZ					2.86			3.18		8.8
					FF/CK		2.81			2.81		6.06
					PAD->FF/D	0.47	0.71		0.48	0.71		
					Setup req.		1.05			1.37		3.75
					Hold req.		-0.59			-0.92		-3.29
	OUT	PDB08DGZ	SDCLK_O	OUT								
					FF/CK		2.81			3.18		4.43
					PAD	4.13	5.42		4.5	5.74		5.94
					Output Delay	1.27	2.56		1.32	2.56		6.98
					Setup margin @ SDRAM							1.51
					Hold margin @ SDRAM		7.44			7.44		7.45
							1.27			1.32		1.51
SDA_O	OUT	PDO08CDG	SDCLK_O	OUT								
					FF/CK		2.81			2.81		4.43
					PAD	4.82	5.23		5.17	5.65		6.9
					Output Delay	1.96	2.37		1.99	2.47		2.47
					Setup margin @ SDRAM							7.53
					Hold margin @ SDRAM		7.63			7.53		2.09
							1.96			1.99		
nS*_O	OUT	PDO08CDG	SDCLK_O	OUT								
					FF/CK		2.81			2.81		4.43
					PAD	4.65	4.92		5	5.25		6.5
					Output Delay	1.79	2.06		1.82	2.07		2.07
					Setup margin @ SDRAM							7.93
					Hold margin @ SDRAM		7.94			7.93		1.92
							1.79			1.82		



System Static Timing Analysis (SSTA)

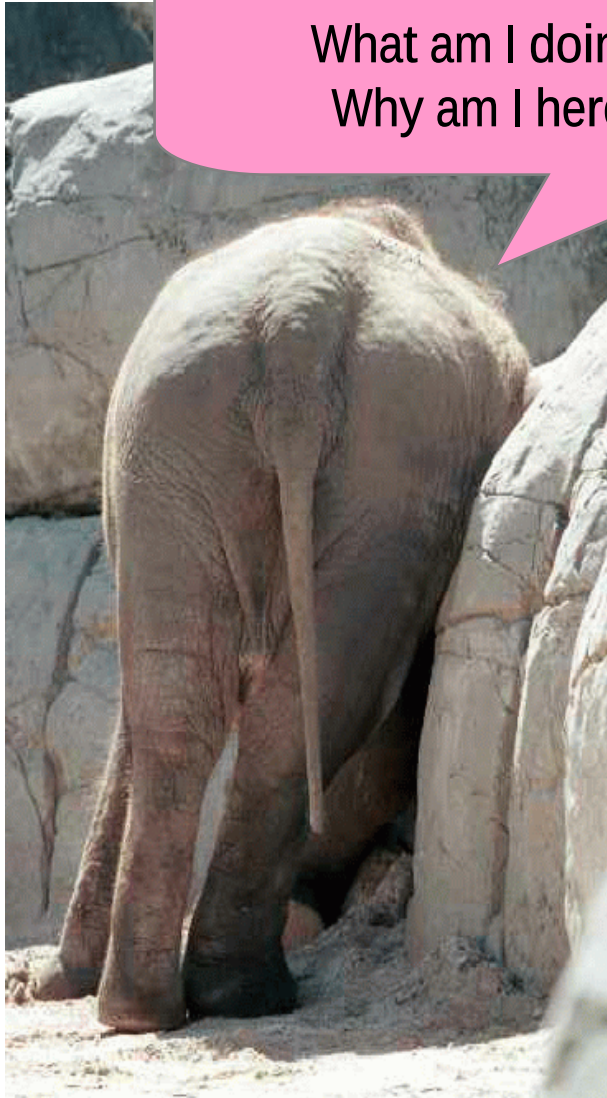


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- **Expand the application of PrimeTime to system perspective**
- **Features:**
 - ◆ **Timing Test Harness**
 - ◆ **Timing Models**
 - ◆ **Run STA**
- **Automate Timing Model generation**

Puzzled Elephant

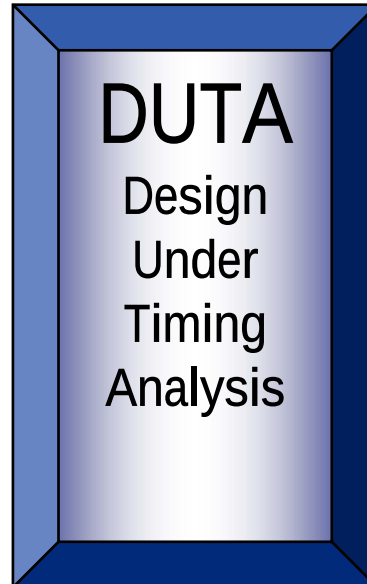
Who am I ?
What am I doing?
Why am I here?



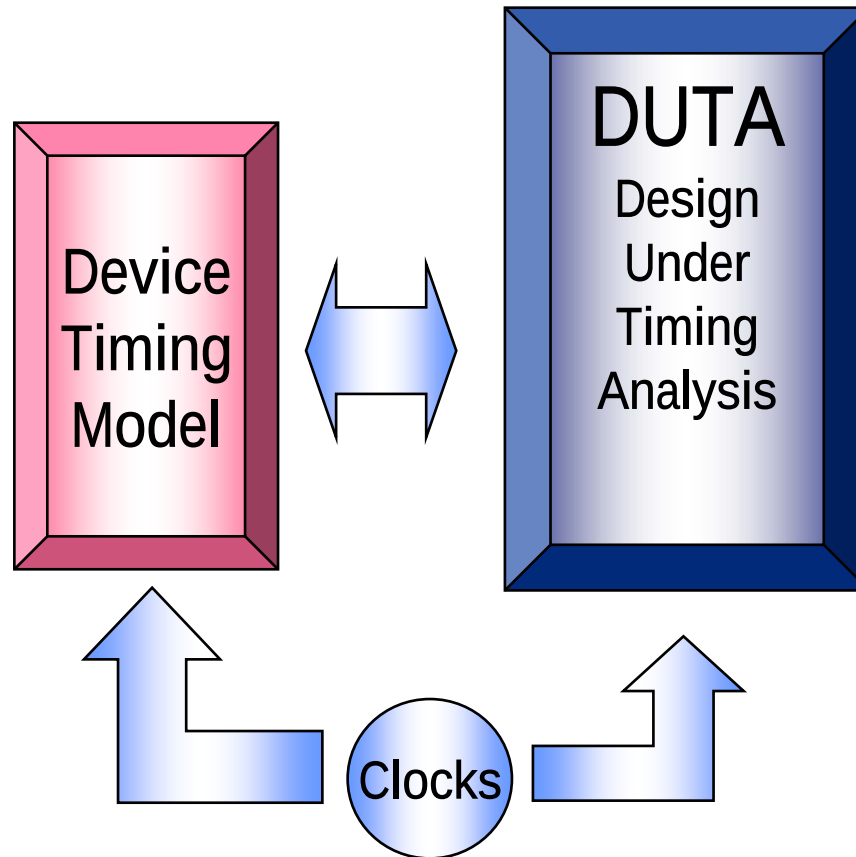
Harness Attached !



Timing Test Harness (TTH)

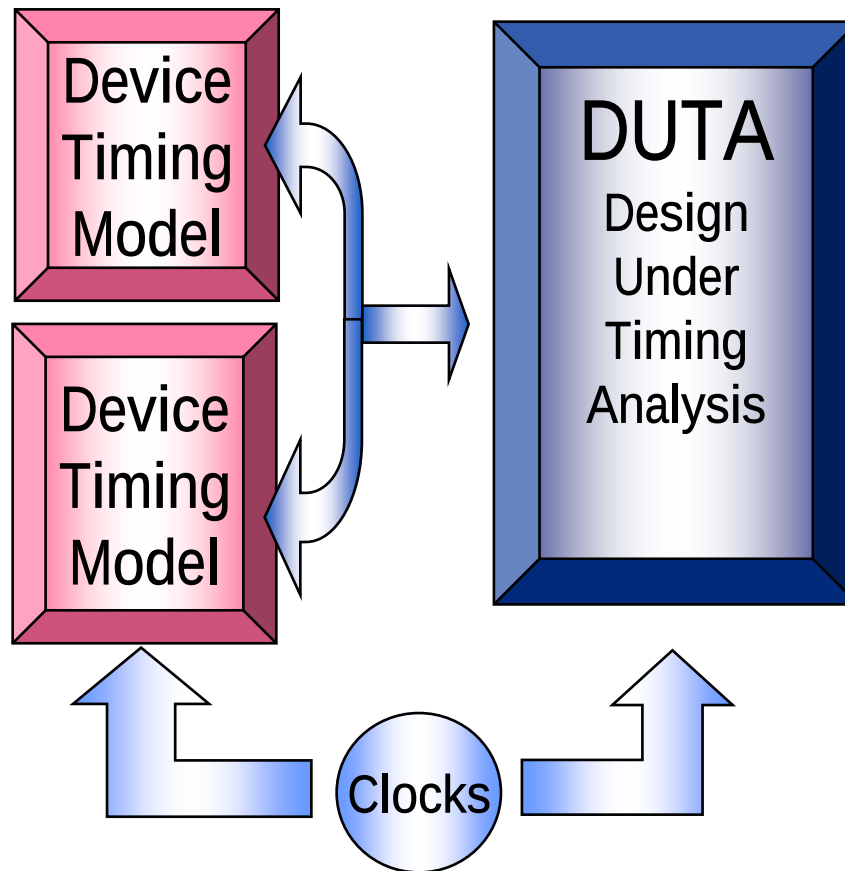


Timing Test Harness (TTH)



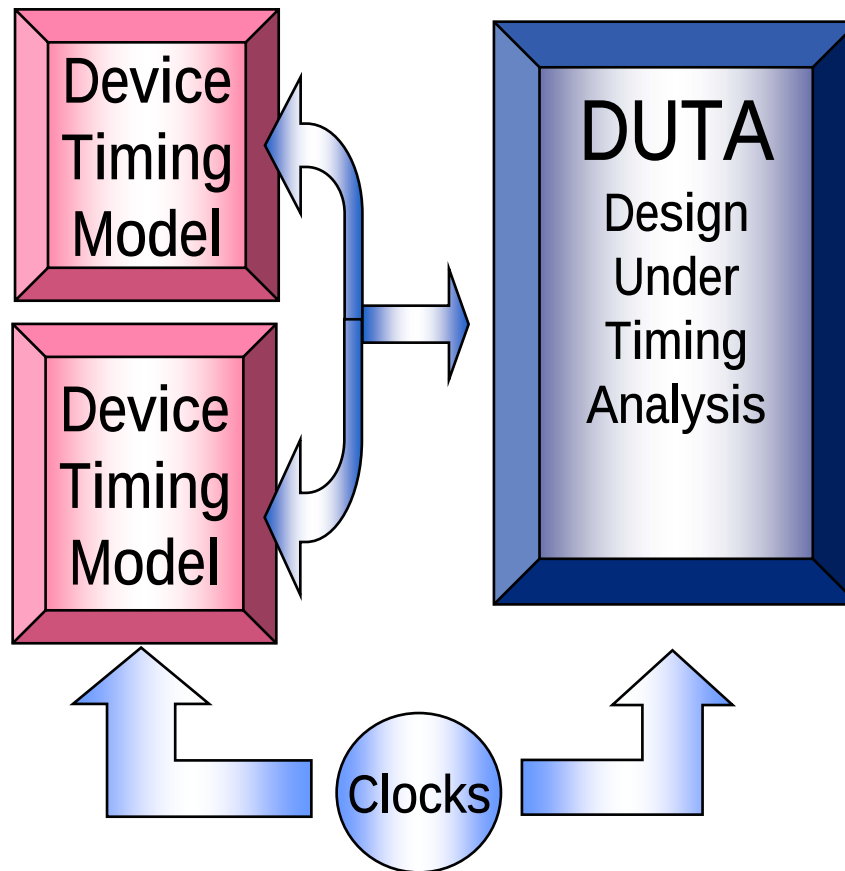
■ **Flexibility**

Timing Test Harness (TTH)



- **Flexibility**
- **Scalability**

Timing Test Harness (TTH)



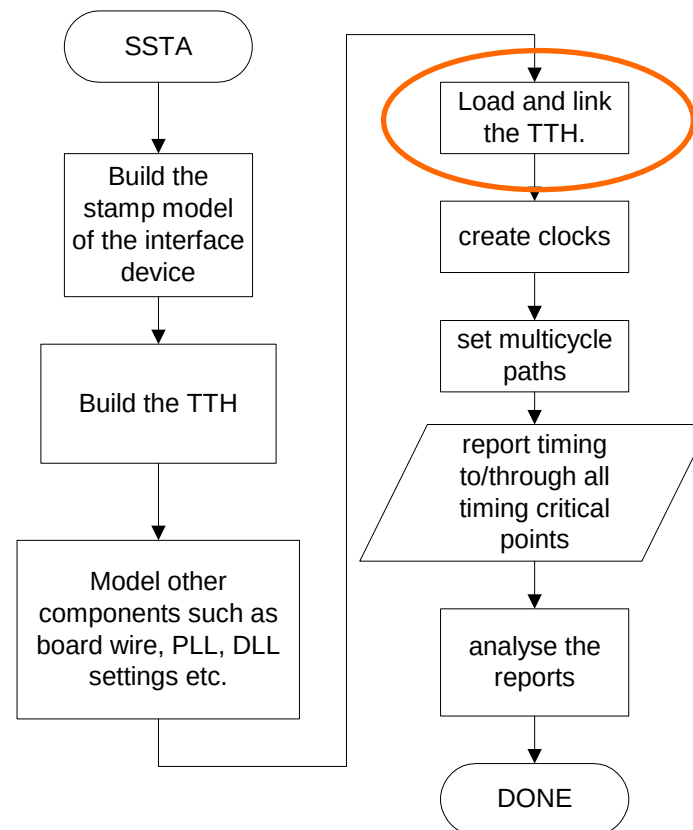
- Flexibility
- Scalability
- Visibility

Stamp Timing Model

- **Model File**
 - ◆ Port name definition
 - ◆ Timing arc declaration
- **Model Data File**
 - ◆ Numerical data of timing arc
 - ◆ Lookup table format
- **Compiled to Synopsys db file by PrimeTime**

Work Flow for SSTA

■ Link the timing test harness





Automation



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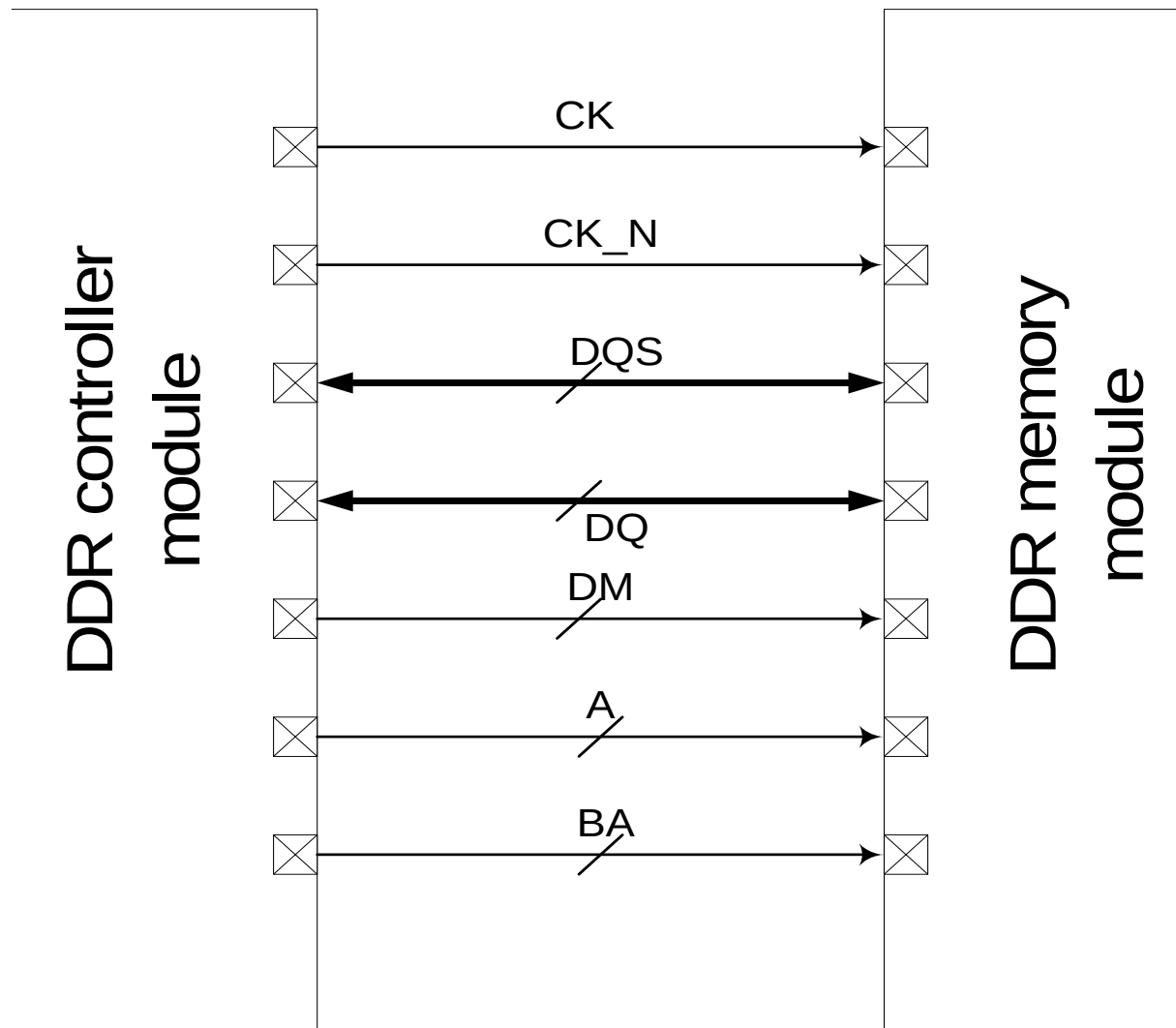
- Stamp model generation
- Different models for DDR SDRAM from different vendors
- One timing test harness for all

System Static Timing Analysis

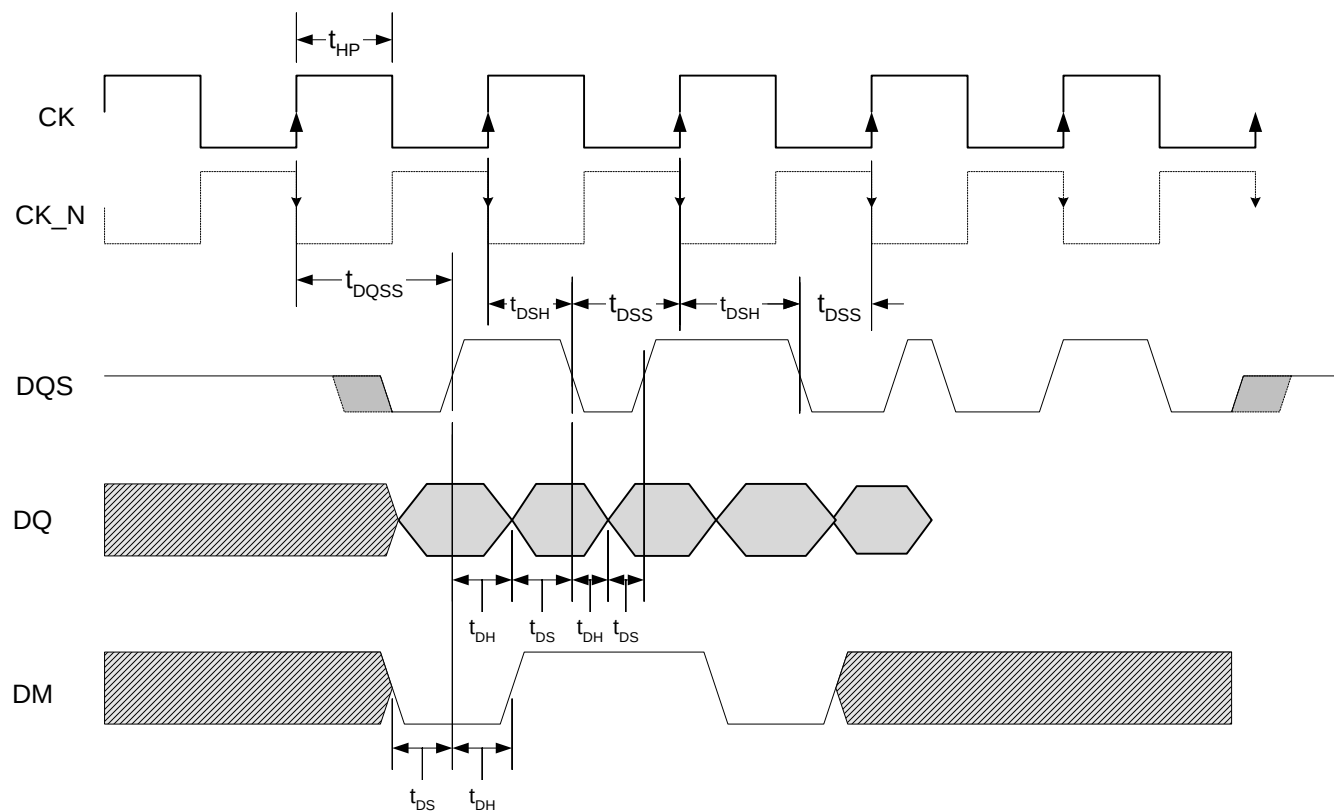
Demonstrate the benefits on a DDR interface

- Challenges
- SSTA on DDR interface

Double Data Rate (DDR) SDRAM Interface

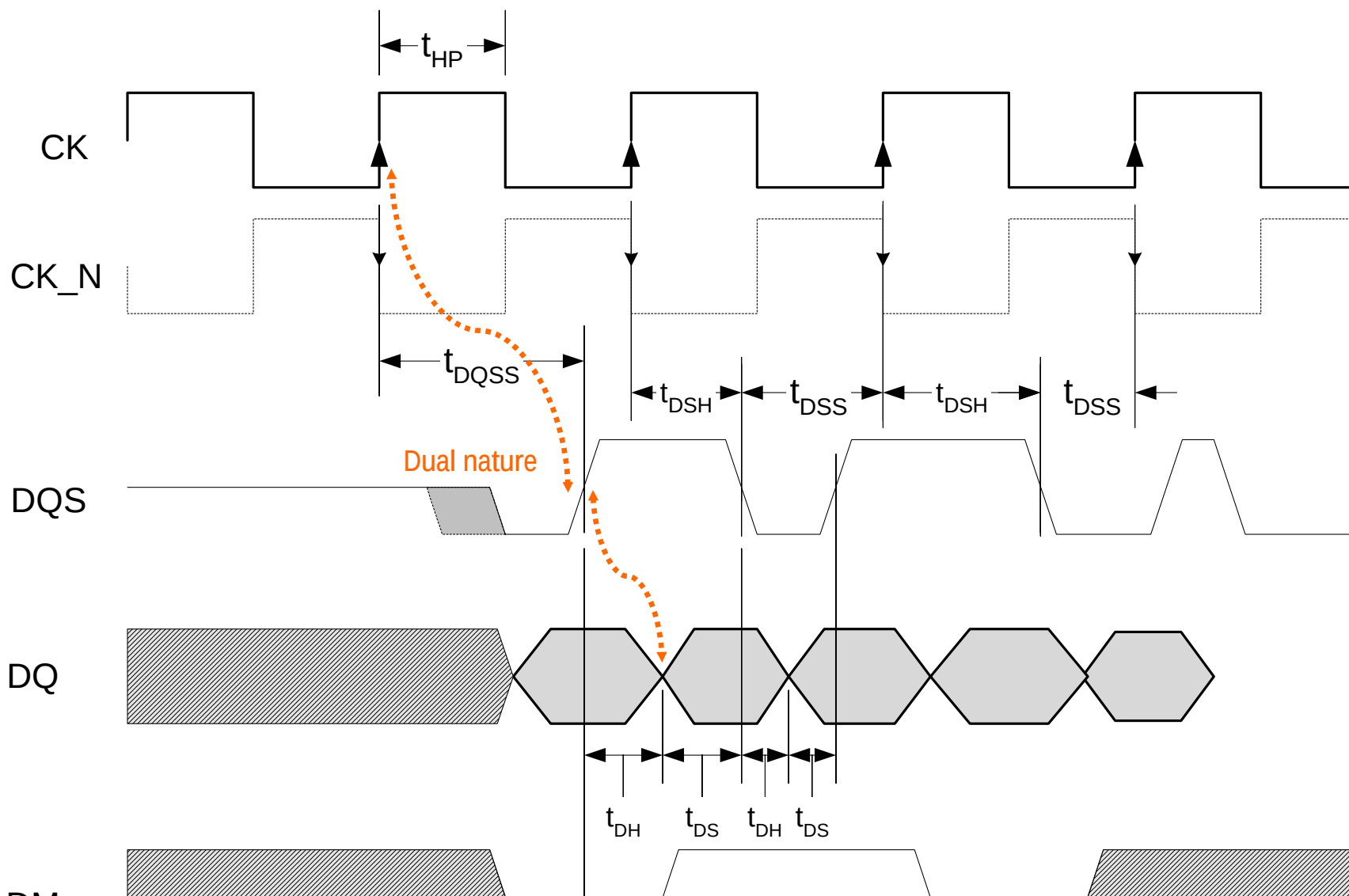


Data Input Timing of DDR SDRAM

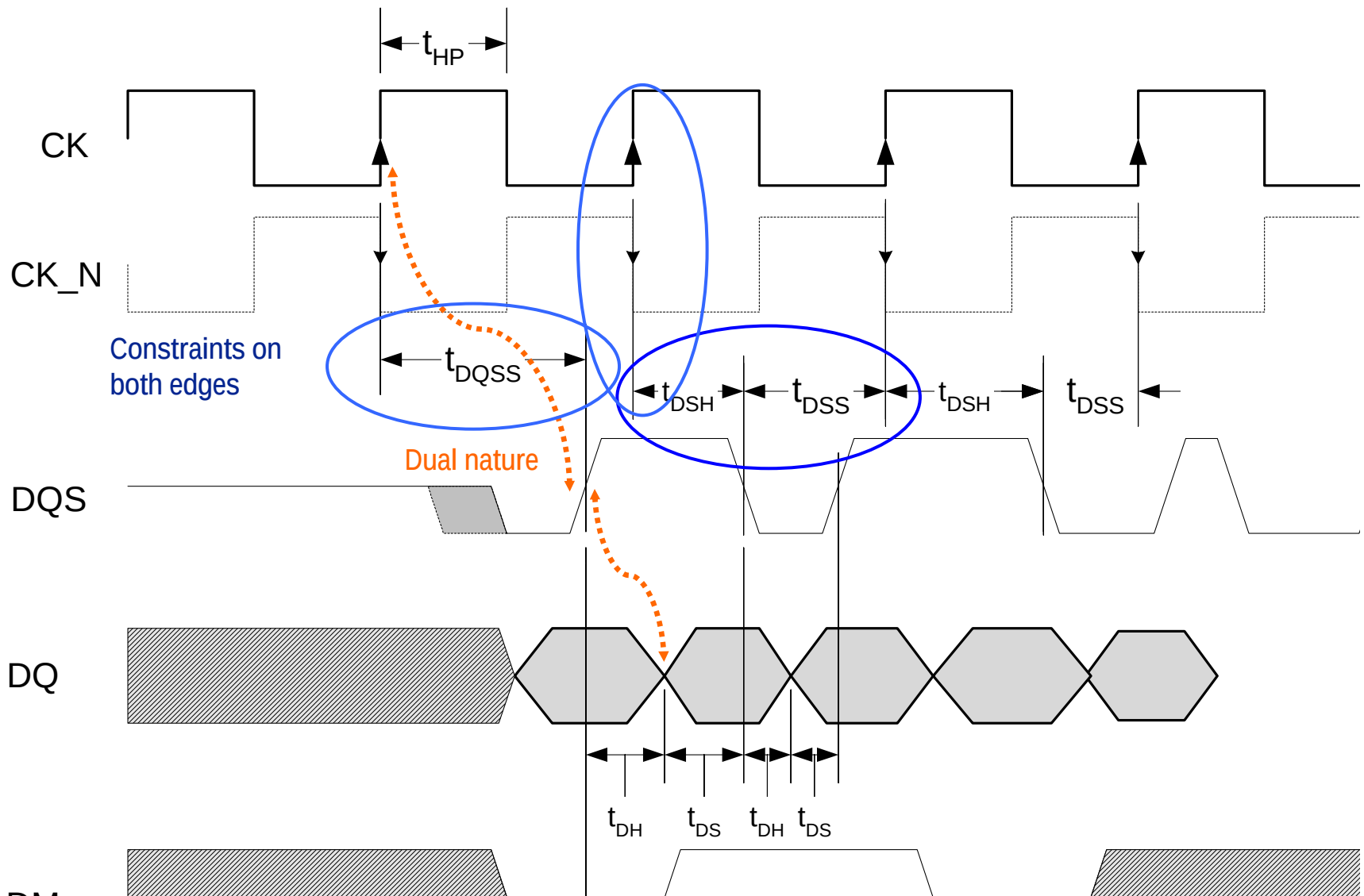


Data Input Timing (w.r.t. DDR SDRAM)

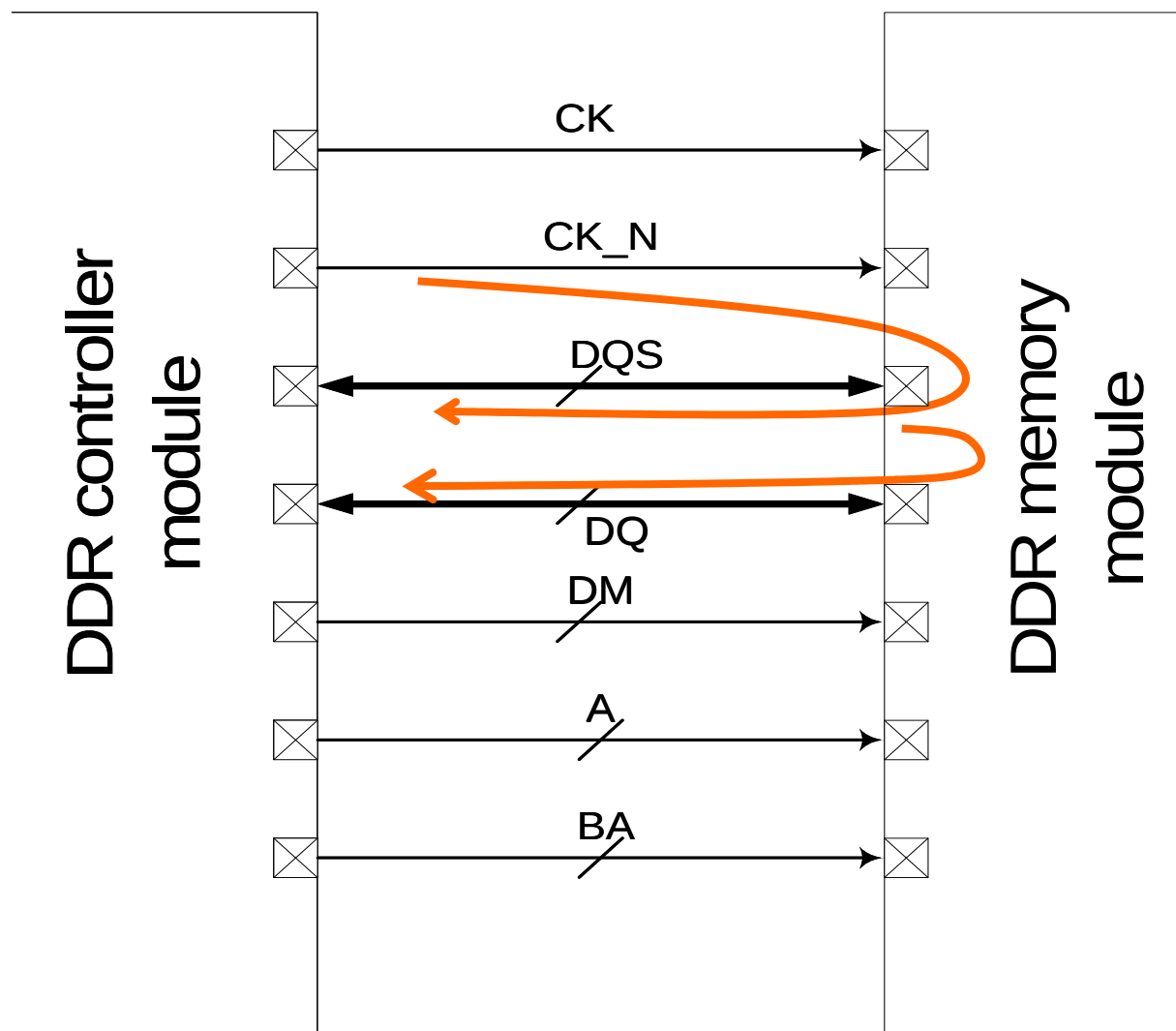
Dual Nature



Constraints On Both Edges



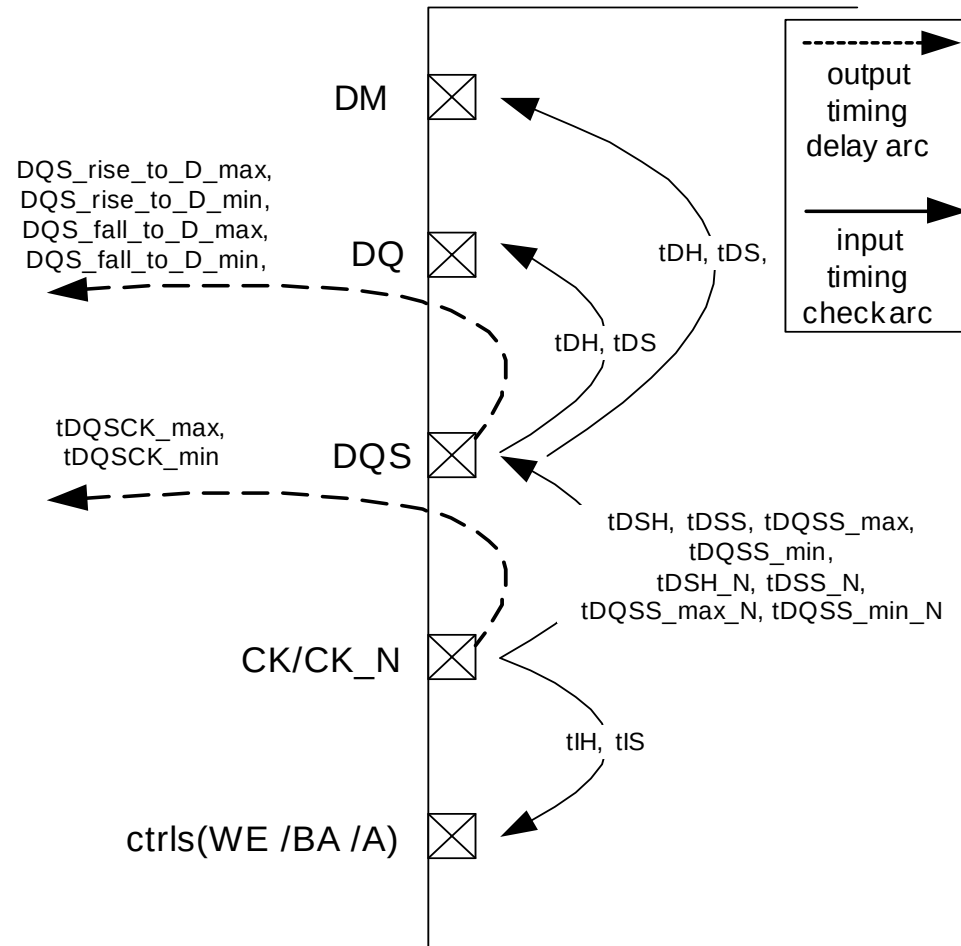
DQS Round Trip Path



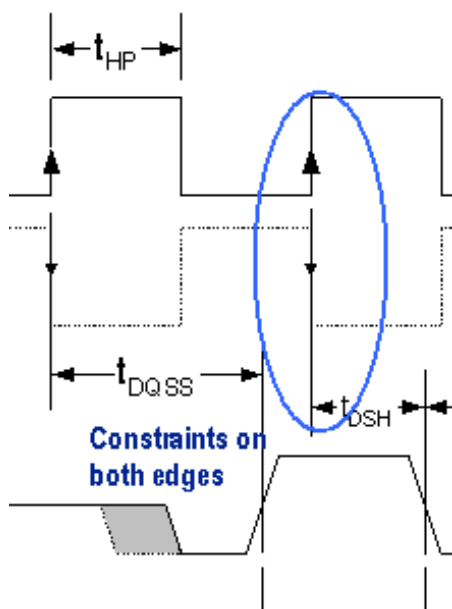
Challenges of STA for DDR

- Dual nature of DQS
- Two edge constraints
 - ◆ Constraints referencing to differential clock
 - ◆ Different constraints for rising and falling edges
- DQS round trip path

DDR SDRAM Stamp Model



Differential Clock Constraints



```

tDSS :
    SETUP (POSEDGE) DQS_in CK;
tDSS_N :
    SETUP (NEGEDGE) DQS_in CK_N;

```

```

ARCDATA
tDSS :
    FALL_CONSTRAINT( constraint_data ) {
        VARIABLE_1 : CONSTRAINED_PIN_TRANSITION
        VARIABLE_2 : RELATED_PIN_TRANSITION
        INDEX_1 ( "0.0000, 1.0000");
        INDEX_2 ( "0.0000, 1.0000");
        VALUES ( "1.6000, 1.6000", "1.6000, 1.6000");
    }
ENDARCDATA

```

```

ARCDATA
tDSS_N :
    RISE_CONSTRAINT( constraint_data ) {
        VARIABLE_1 : CONSTRAINED_PIN_TRANSITION
        VARIABLE_2 : RELATED_PIN_TRANSITION
        INDEX_1 ( "0.0000, 1.0000");
        INDEX_2 ( "0.0000, 1.0000");
        VALUES ( "1.6000, 1.6000", "1.6000, 1.6000");
    }
ENDARCDATA

```

Different Constraints for Different Edges

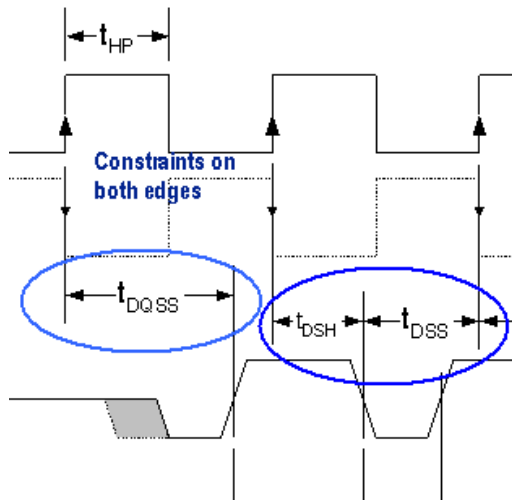
```
tDSS :
    SETUP (POSEDGE) DQS_in CK;
tDQSS_max :
    SETUP (NEGEDGE) DQS_in CK;
```

ARCDATA

```
tDSS :
    FALL_CONSTRAINT( constraint_data ) {
        VARIABLE_1 : CONSTRAINED_PIN_TRANSITION
        VARIABLE_2 : RELATED_PIN_TRANSITION
        INDEX_1 ( "0.0000, 1.0000");
        INDEX_2 ( "0.0000, 1.0000");
        VALUES ( "1.6000, 1.6000", "1.6000, 1.6000");
    }
ENDARCDATA
```

ARCDATA

```
tDQSS_max :
    RISE_CONSTRAINT( constraint_data ) {
        VARIABLE_1 : CONSTRAINED_PIN_TRANSITION
        VARIABLE_2 : RELATED_PIN_TRANSITION
        INDEX_1 ( "0.0000, 1.0000");
        INDEX_2 ( "0.0000, 1.0000");
        VALUES ( "2.0000, 2.0000", "2.0000, 2.0000");
    }
ENDARCDATA
```



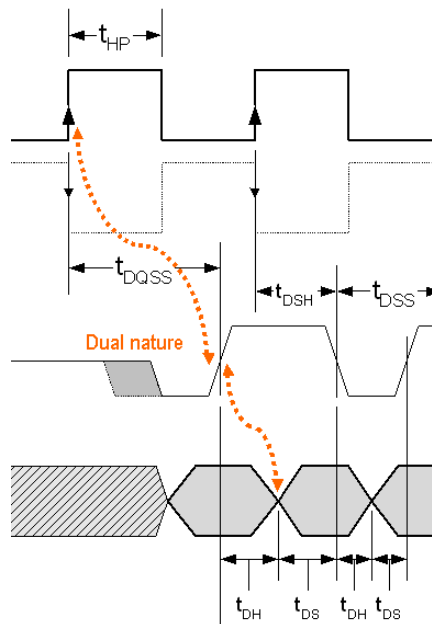
Dual Nature of DQS

DQS_fall_to_D_max :

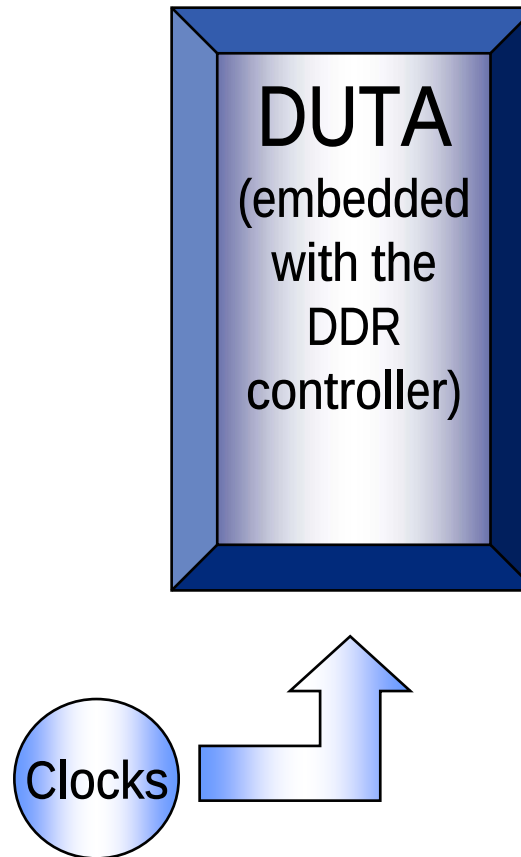
```
DELAY (NEGEDGE) DQS_out { D_out[0], D_out[1], D_out[2],
D_out[3], D_out[4], D_out[5], D_out[6], D_out[7], D_out[8],
D_out[9], D_out[10], D_out[11], D_out[12], D_out[13],
D_out[14], D_out[15], D_out[16], D_out[17], D_out[18],
D_out[19], D_out[20], D_out[21], D_out[22], D_out[23],
D_out[24], D_out[25], D_out[26], D_out[27], D_out[28],
D_out[29], D_out[30], D_out[31]};
```

tDQSK_max :

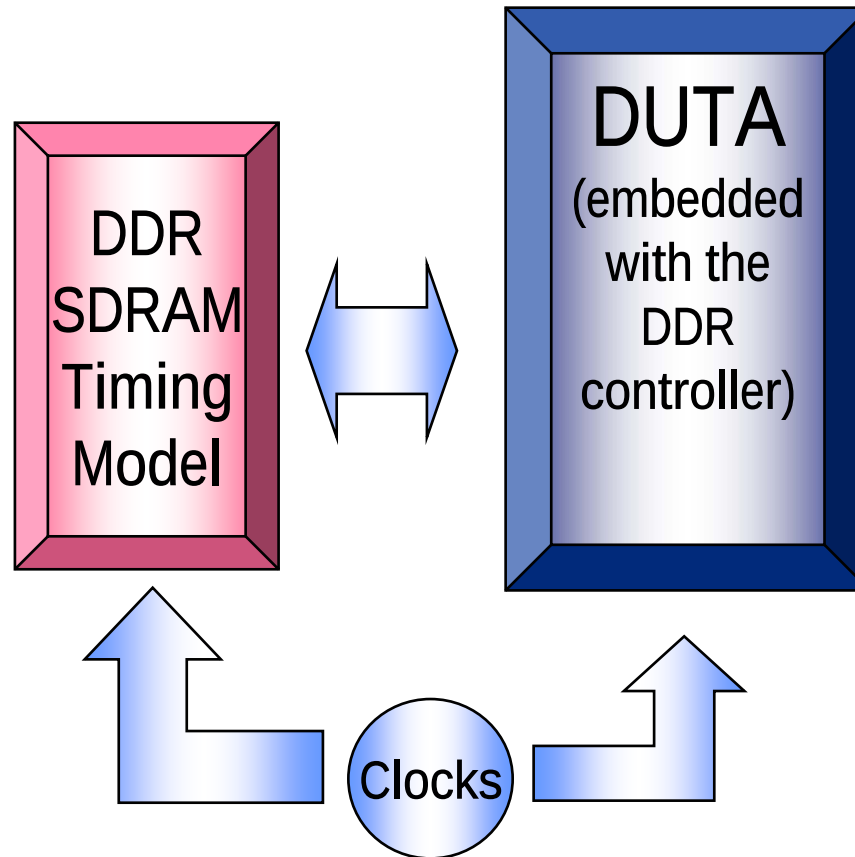
```
DELAY (NONINVERTING) CK DQS_out;
```



TTH for a DDR Interface

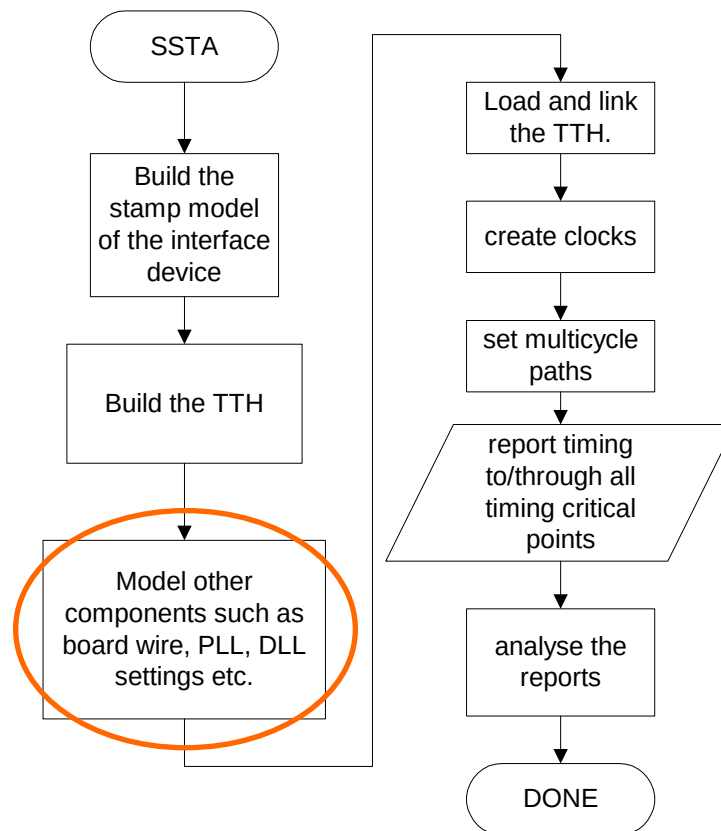


TTH for a DDR Interface



Work Flow for SSTA on a DDR Interface

■ Model PLL and DLL





Report Sample



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Startpoint: chip_inst/chip_top_inst/ddr_control_inst/io/dqs_0/dqs_oe_int_reg
(rising edge-triggered flip-flop clocked by hclk)
Endpoint: DRAM (falling edge-triggered flip-flop clocked by DRAM_CLK)
Path Group: DRAM_CLK
Path Type: max
Point Incr Path

clock hclk (rise edge)	0.00	0.00
clock network delay (propagated)	6.48	6.48
chip_inst/chip_top_inst/ddr_control_inst/io/dqs_0/dqs_oe_int_reg/CK(SMDFFHQX2)	0.00	6.48 r
chip_inst/chip_top_inst/ddr_control_inst/io/dqs_0/dqs_oe_int_reg/Q(SMDFFHQX2)	0.25	6.73 r
chip_inst/chip_top_inst/ddr_control_inst/io/dqs_0/dqs_oe (io_dqscell)	0.00	6.73 r
chip_inst/chip_top_inst/ddr_control_inst/io/dqs_oe[0] (io)	0.00	6.73 r
chip_inst/chip_top_inst/ddr_control_inst/dqs_oe[0] (ddr_control)	0.00	6.73 r
chip_inst/chip_top_inst/DDRDQS_E[0] (obelix_top)	0.00	6.73 r
chip_inst/DQS_B_0_buf/PAD (sstl)	2.66	9.39 r
chip_inst/DQS_B_0 (chip)	<- 0.50 *	9.89 r
DRAM/DQS_in (mt46v16M4)	0.00 *	9.89 r
data arrival time		9.89
clock DRAM_CLK (fall edge)	12.00	12.00
clock network delay (propagated)	-0.64	11.36
clock reconvergence pessimism	0.00	11.36
DRAM/CK (mt46v16M4)		11.36 f
library setup time	-2.00	9.36
data required time		9.36

data required time		9.36
data arrival time		-9.89

slack (VIOLATED)		-0.53

Conclusion

- **System static timing analysis (SSTA)**
- **Timing test harness (TTH)**
- **Automation of Stamp model generation**
- **Model DDR completely by Stamp model**
 - ◆ **DQS challenges**
- **Visibility from the system perspective**



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System Static Timing Analysis

Questions



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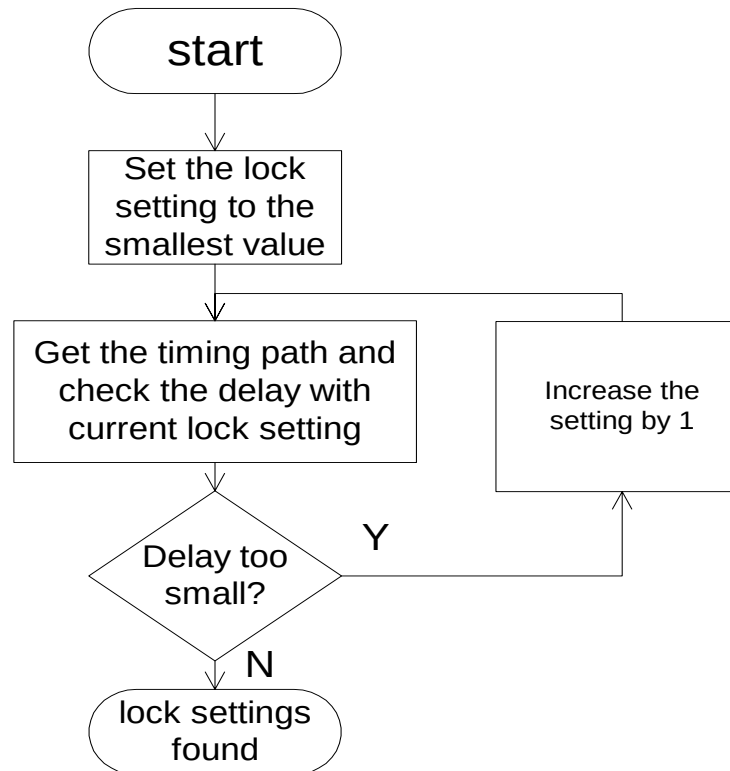
System Static Timing Analysis

Questions

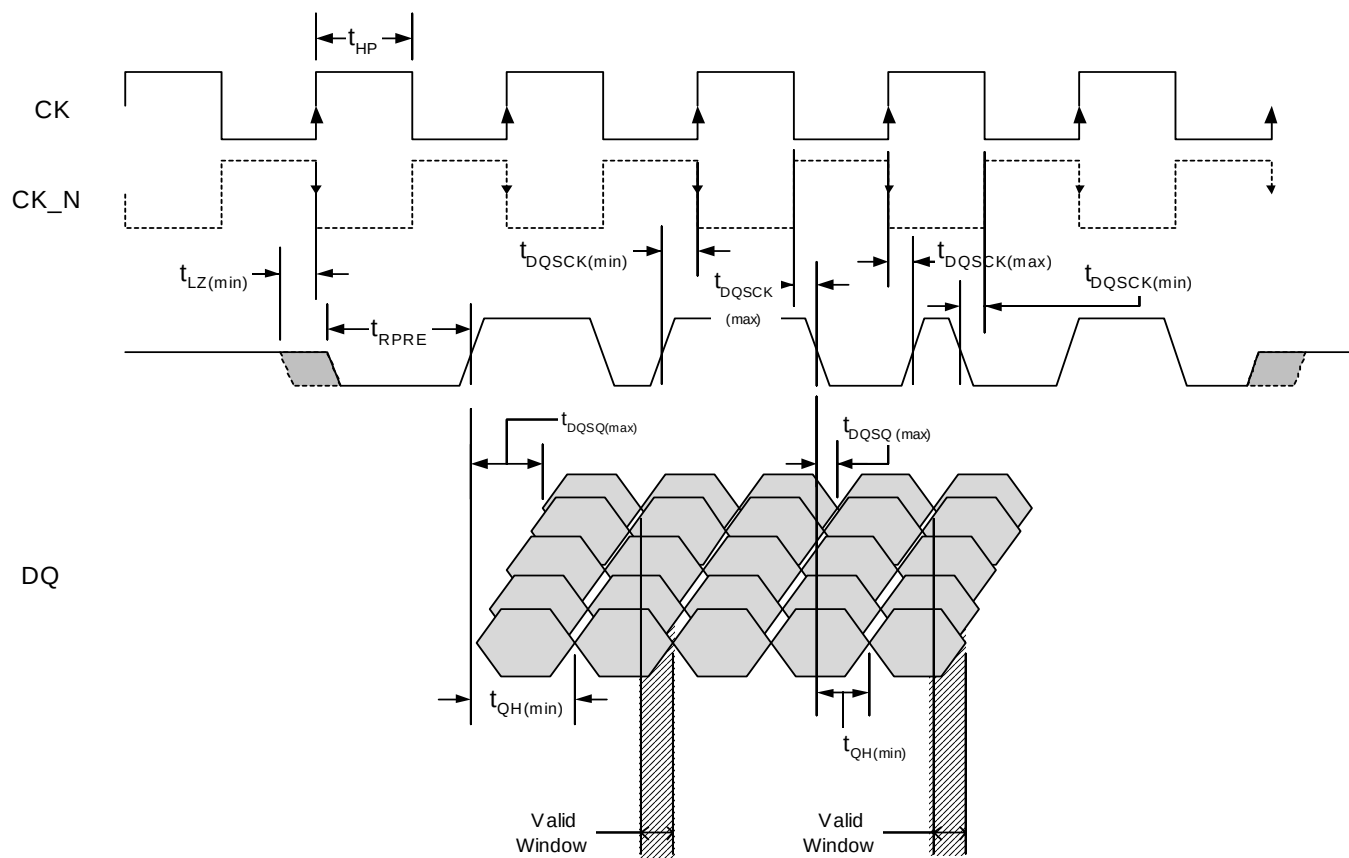
System Static Timing Analysis

Questions

DLL Locking in PrimeTime

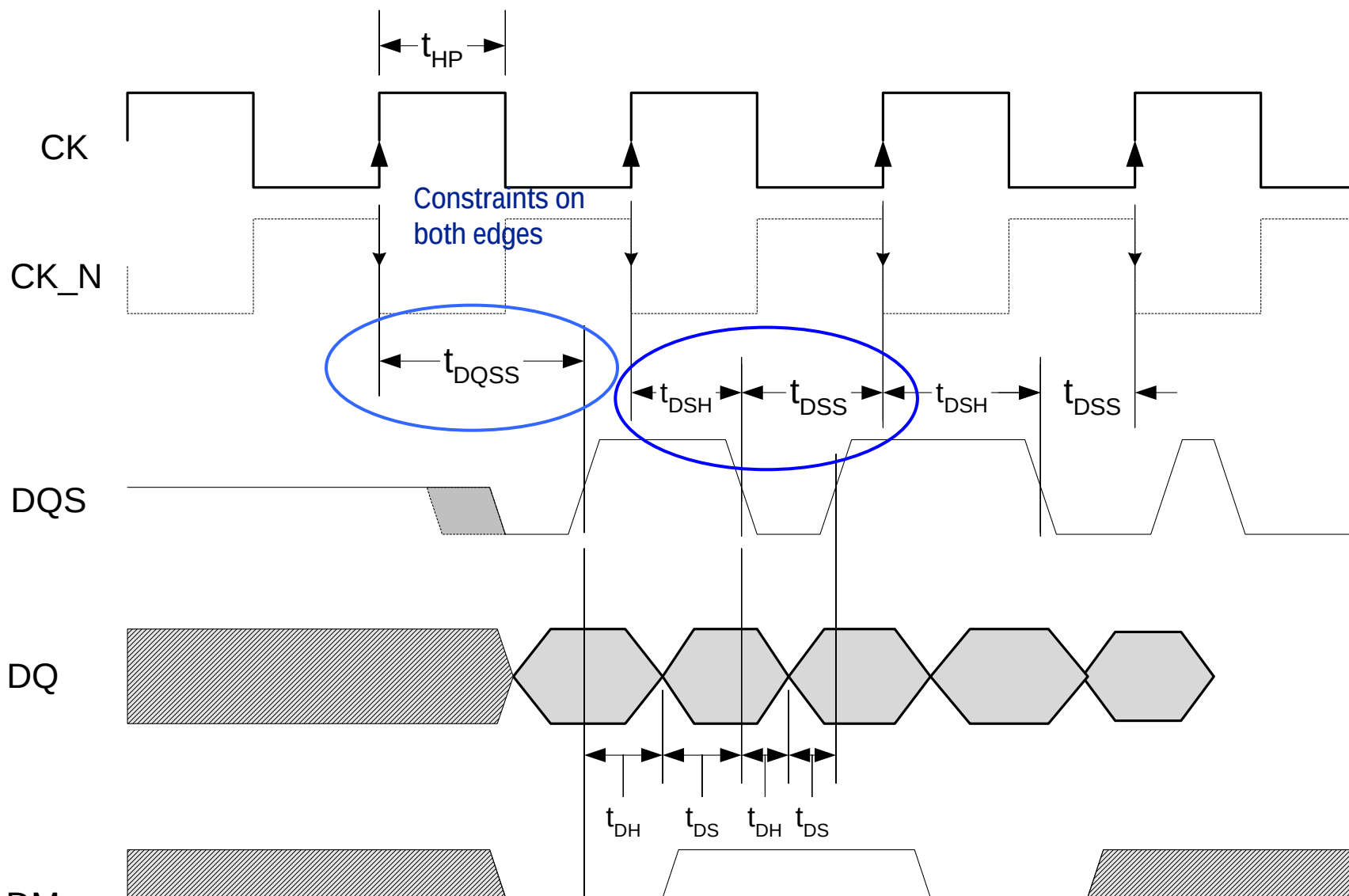


DDR Output Timing Requirement



Data Output Timing (w.r.t. DDR SDRAM)

Constraints On Both Edges



Constraints On Both Edges

